

Global Infrastructure & Networking

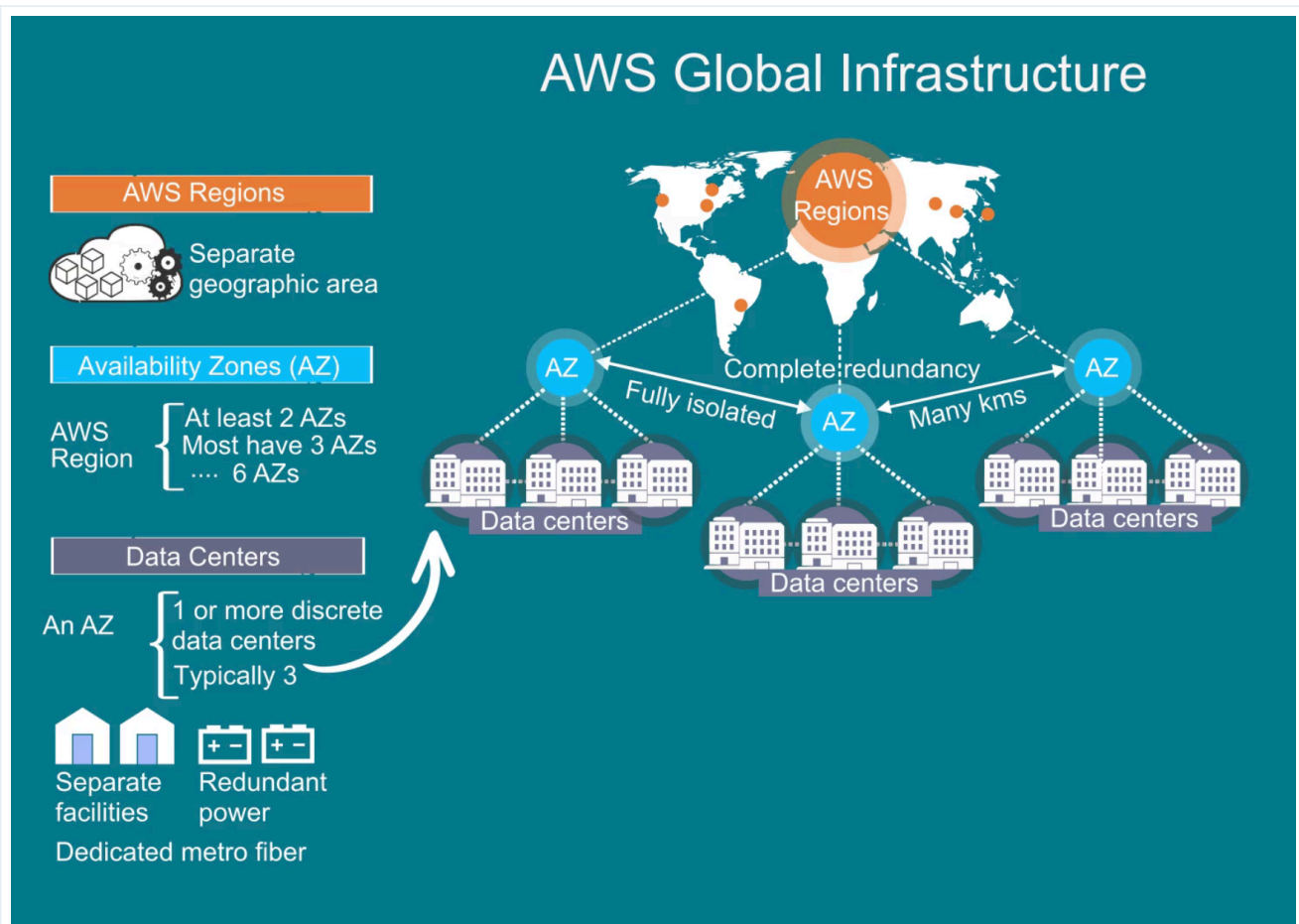
A practical continuation of the AWS foundations reference

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01	Networking concepts VPCs, subnets, addressing, routing and traffic control.
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AWS Global Infrastructure recap

AWS is not one giant data center. It is a worldwide infrastructure made of separate geographic Regions, with isolated Availability Zones inside them and global networking that connects services and customers.



RECAP

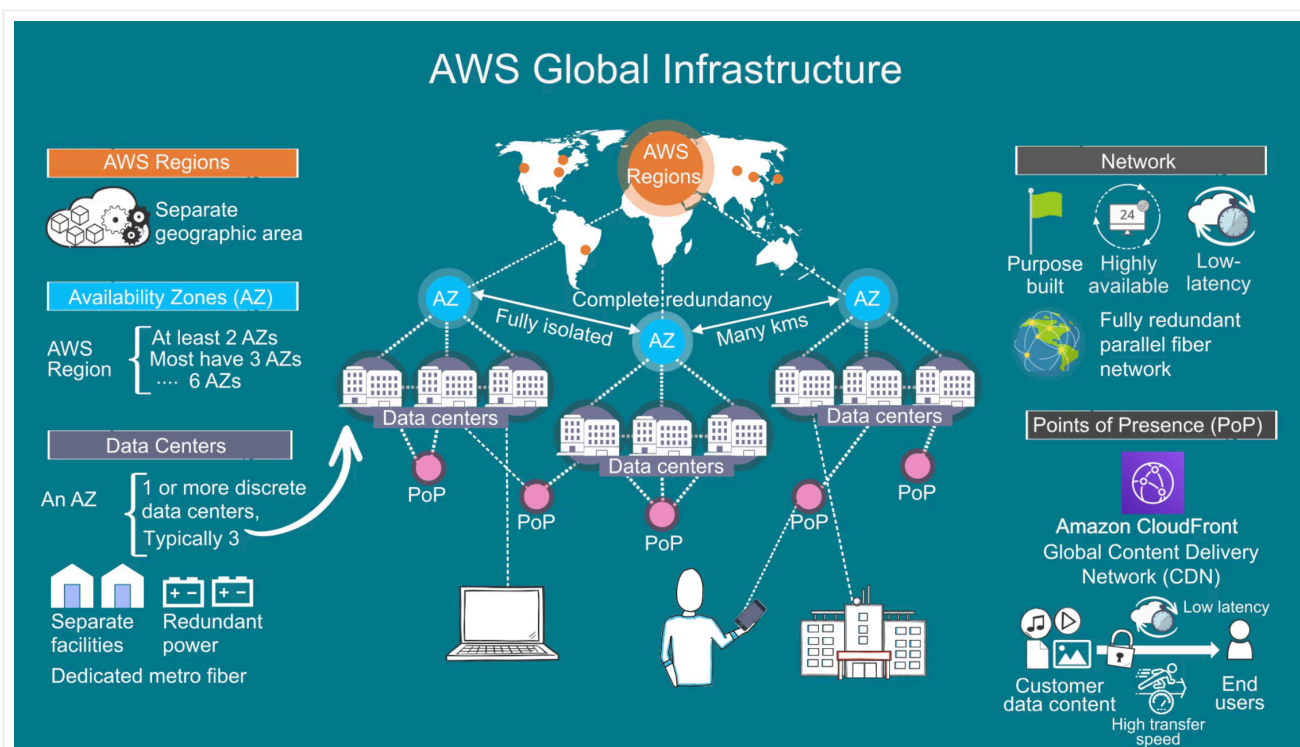
Region, AZ and data centers

A Region is a separate geographic AWS area. Inside it are multiple Availability Zones.

- An AZ is an isolated location made up of one or more discrete data centers.
- AZs are separated for fault isolation, but connected through high-speed, low-latency networking.
- AWS continues to add Regions, AZs and other infrastructure as its global footprint grows.

Points of Presence and global delivery

Regions and Availability Zones are where workloads run, but customers do not always need to connect directly to a Region. AWS also operates Points of Presence (PoPs) around the world. Services such as CloudFront use this edge infrastructure to deliver content closer to end users.



EDGE DELIVERY

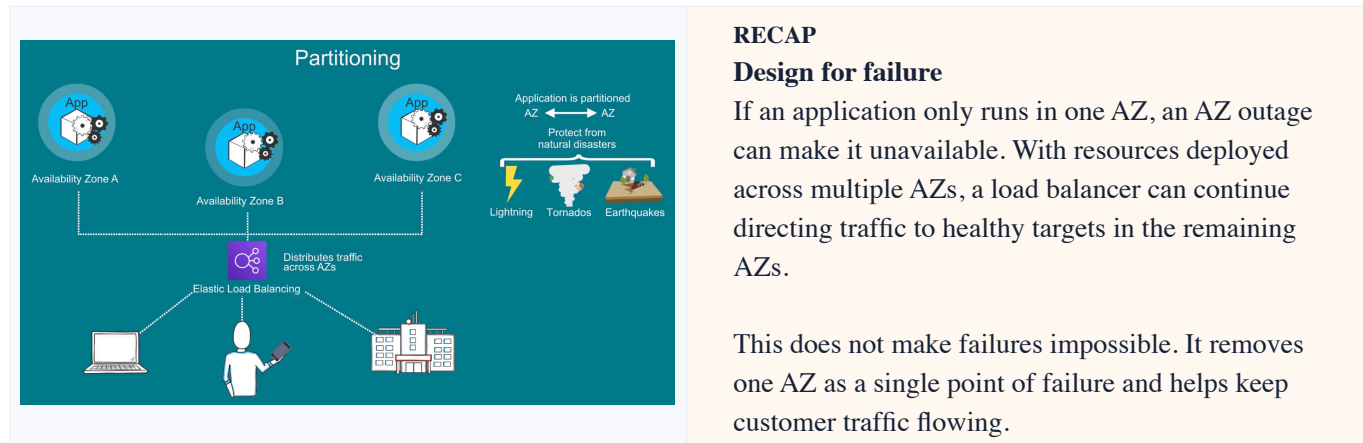
PoPs bring content closer to users

- A Point of Presence is an AWS edge location used by services such as CloudFront.
- CloudFront can cache copies of content at edge locations, reducing the distance many user requests need to travel.
- On a cache miss, CloudFront retrieves the content from the configured origin, such as S3 or an application, and can cache the response for later requests.
- AWS connects its infrastructure through a highly available, redundant global network, helping traffic move between edge locations, Regions and AWS services.

Mental model: Region/AZ = where your workload runs. PoP/edge = where services such as CloudFront can meet users closer to them.

Why multiple Availability Zones matter

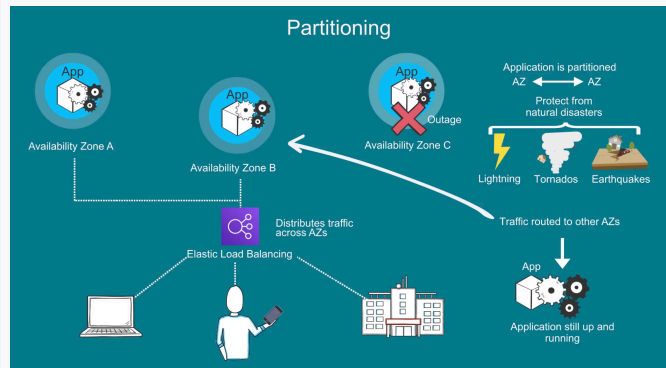
This is the reliability idea from Part I: do not make one Availability Zone the only place your application can run. Spread the workload across AZs, then use load balancing to send traffic to healthy resources.



WHEN AN AZ BECOMES UNAVAILABLE

Traffic should have somewhere else to go

The second diagram shows the failure case. If one AZ becomes unavailable, traffic can be routed to application resources running in other AZs. This is the practical reason AWS Regions contain isolated Availability Zones.



That gives us the bridge into networking: AWS can provide the global infrastructure and isolated locations, but our network design decides how customers actually reach the healthy parts of the application.

What the global infrastructure gives us

AWS can place infrastructure closer to customers, spread workloads across isolated failure domains and adjust capacity as needs change. The result is a platform designed around reach, reliability, performance and flexibility without owning the physical infrastructure yourself.

In short: choose where resources run, scale them when needed, and use AWS networking to reach users globally.

t3.micro	m4.large
0.0104 USD per Hour	0.1000 USD per Hour
0.0196 USD per Hour	0.1920 USD per Hour
2 (1 core)	2 (1 core)
1024	8192
–	–
ebs	ebs
Up to 5 Gigabit	Moderate

ALREADY SEEN IN PRACTICE

Flexibility at the compute level

In Part I we stopped an EC2 instance, reviewed a different compatible instance type and could replace its compute capacity with a larger option.

That is one small example of the wider cloud idea: infrastructure is not fixed hardware you bought years ago. It can be adjusted as the workload changes.

Important distinction

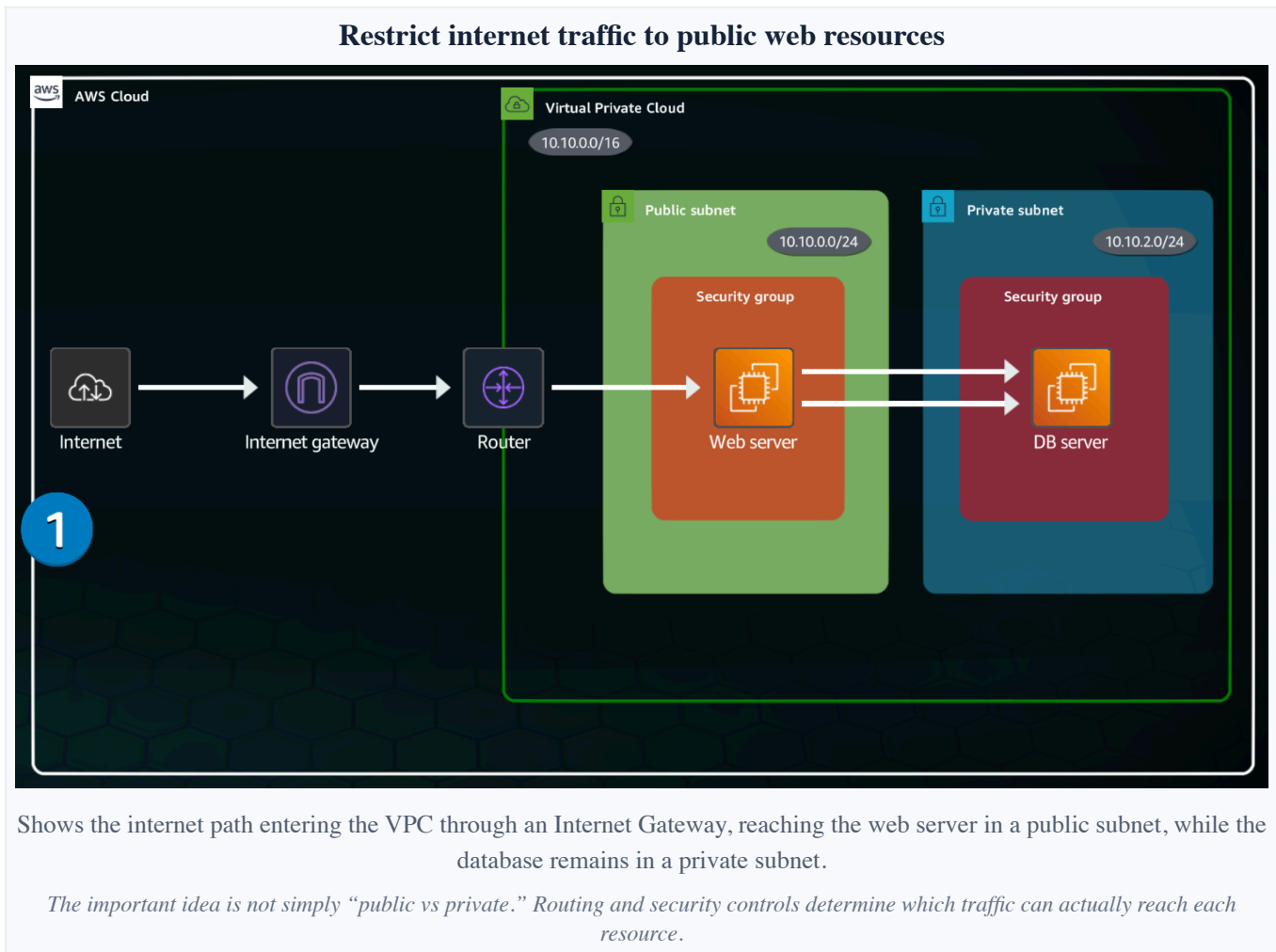
AWS expanding its footprint does not mean your resources automatically move into new Regions or AZs. You choose where your workload is deployed. The expanding footprint gives you more placement options over time.

01 Networking Concepts

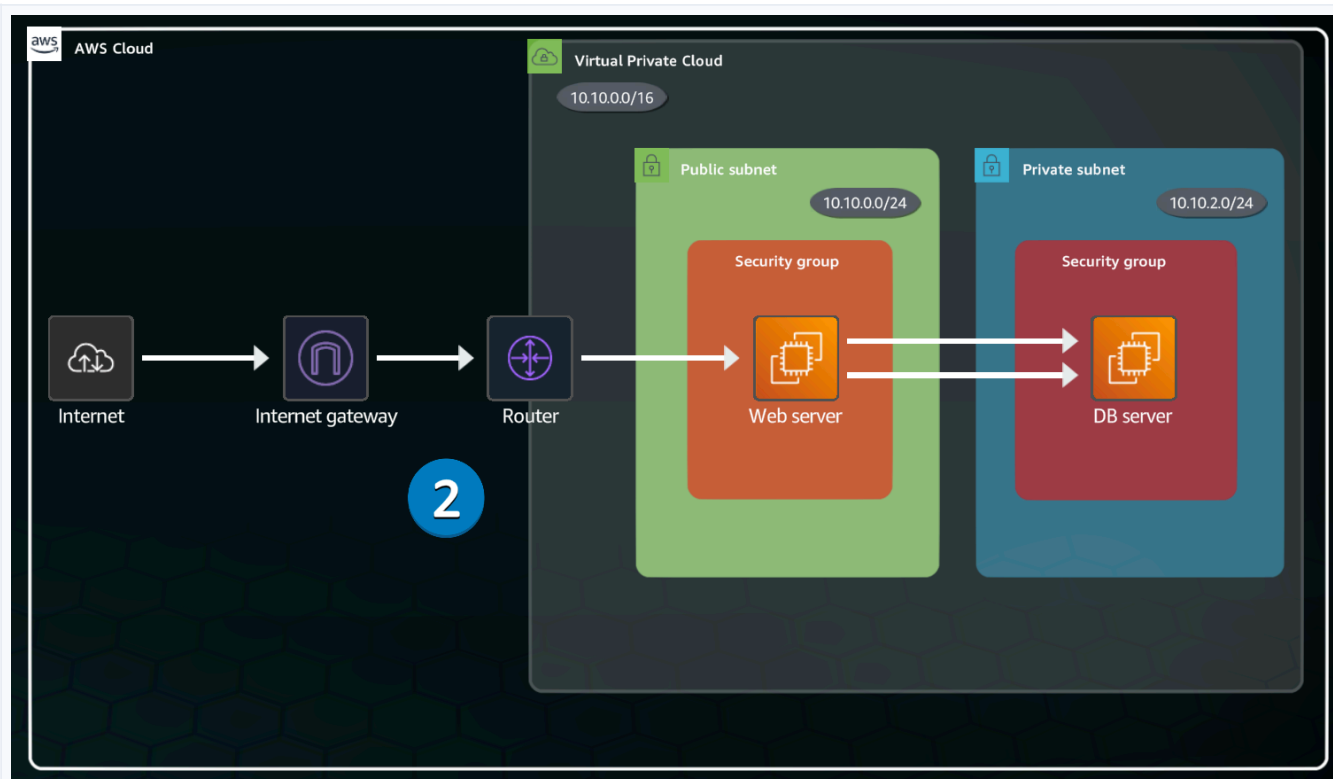
Now that we know where AWS resources can run, the next question is how they communicate. We start with the VPC, then narrow the network into subnets before looking at routes, addresses and traffic rules.

From internet traffic to a VPC

The first networking idea is separation. A customer-facing web server may need a path from the internet, while a database usually should not. AWS networking lets us decide which resources are exposed and which remain private.



The architecture keeps the web server and database inside the same VPC, but gives them different network roles. The web tier can accept the traffic it needs from outside; the database can be reached by the application without being directly exposed to the internet.



The application network inside a VPC

File: *networking-concept-restriction-02.png*

Focuses on the VPC boundary around the application. Public and private subnets divide the network, while Security Groups control allowed traffic to the web and database servers.

What is a VPC?

Imagine you want to run a web server and a database in AWS.

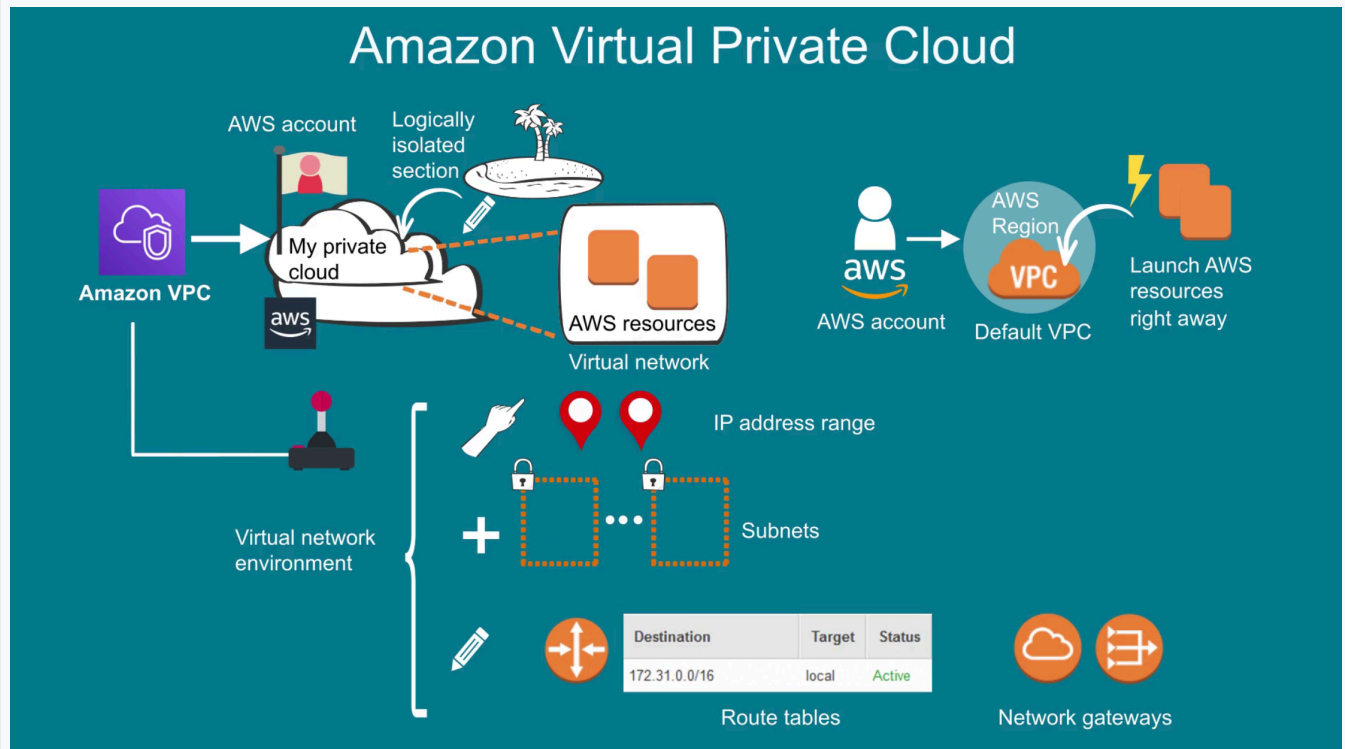
How do you decide which resources can reach the internet, which can communicate with each other, and which should stay private?

You could manage each resource individually, but that quickly becomes difficult as the application grows. A VPC gives you one network environment in which to organise and control all of this.

A Virtual Private Cloud (VPC) is a logically isolated virtual network in one AWS Region. It gives your AWS resources an IP address range and provides the network structure they use to communicate. You then divide that range into subnets, define routes, attach gateways when needed, and control traffic with security rules.

AWS provides a default VPC in each Region so resources can be launched quickly, but you can also create your own VPC when you want control over the address range and network design.

Amazon VPC overview



A VPC is regional, but a subnet is not. Each subnet exists entirely inside one Availability Zone. This is why a VPC can contain separate subnets in multiple AZs while still acting as one regional network.

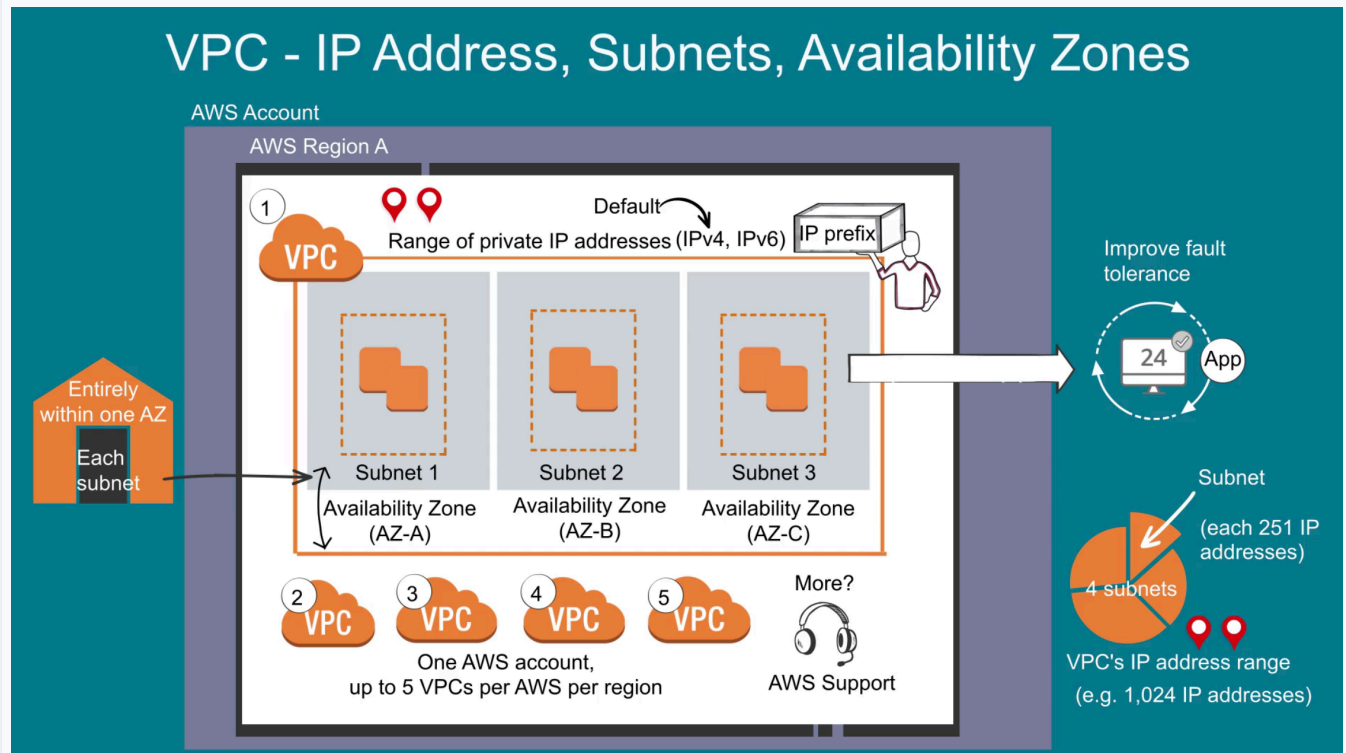
Suppose you want your application to keep running if one Availability Zone has a problem.

Can you simply create one subnet and spread its resources across multiple Availability Zones?

No. A subnet exists entirely inside one AZ.

The VPC is regional, so it can contain multiple subnets in different Availability Zones.

VPC, IP ranges, subnets and Availability Zones



Shows one VPC IP range divided into subnets across different Availability Zones. Each subnet receives part of the VPC address space and belongs to exactly one AZ.

We know that a VPC can contain several subnets. But if they all belong to the same VPC, how does each subnet get its own part of the network?

This is where IP ranges come in.

The VPC starts with one larger range of IP addresses. When you create a subnet, you give it a smaller range taken from the VPC's range. For example:

VPC: 10.0.0.0/16

Subnet A - 10.0.1.0/24 (AZ-A)

Subnet B - 10.0.2.0/24 (AZ-B)

So there are really two things happening at once: Availability Zone is where the subnet exists. Ip Range is which addresses belong to the subnet.

A subnet defines where your resources run and which IP addresses they can use. This is important because a subnet determines where your resources can run and which network they belong to, helping you organise resources across Availability Zones and control how traffic moves between them.

Public and private traffic

Concept	Beginner mental model	What matters
Public subnet	Can have internet connectivity	Route to an Internet Gateway
Private subnet	No direct inbound internet path	No direct route from the internet
Security Group	Resource firewall	Controls allowed traffic

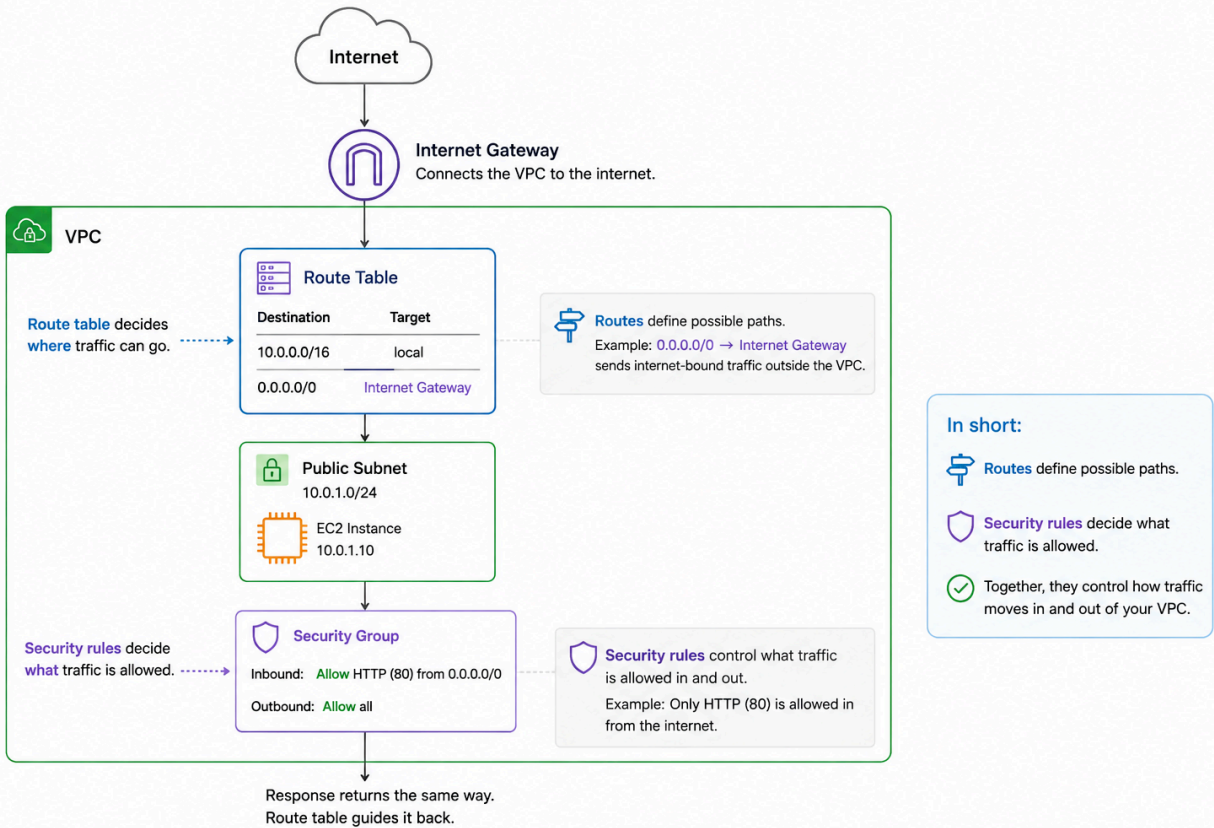
TRAFFIC PATH

Routes and gateways

A route table tells traffic where it can go. An Internet Gateway connects a VPC to the internet when routing and security allow it.

Routes define possible paths.

Security rules decide what traffic is allowed.



Private IP, public IP and DNS

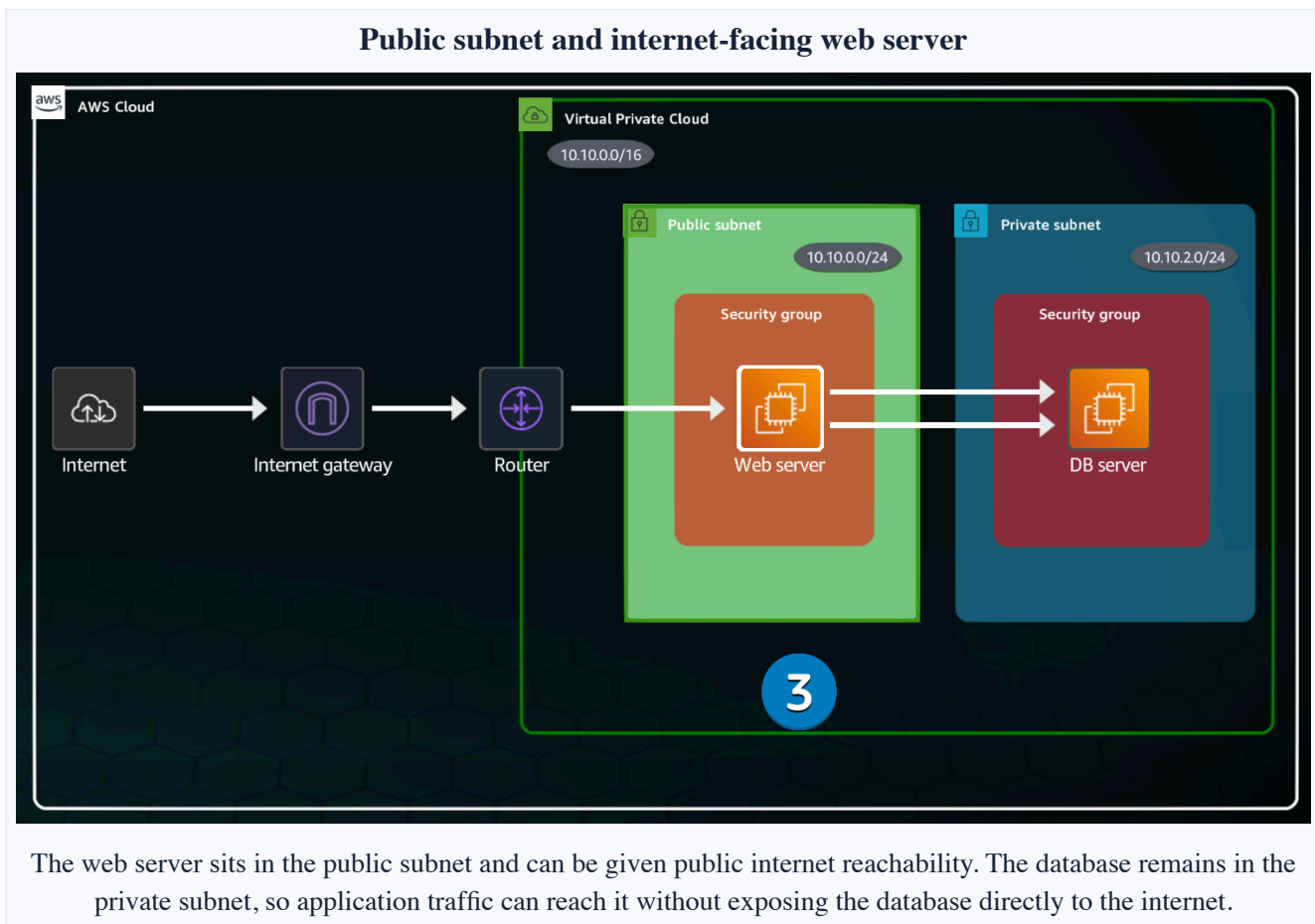
Private IP: Address used inside the VPC.

Public IP: Internet-routable address.

DNS: Human-readable name that resolves to an address.

Subnets, CIDR blocks and routing

Now that the public and private traffic model is clear, the next step is to see how AWS creates those network boundaries. A subnet is a smaller IP address range inside a VPC, and its routes help determine where traffic can go.



How the network is divided

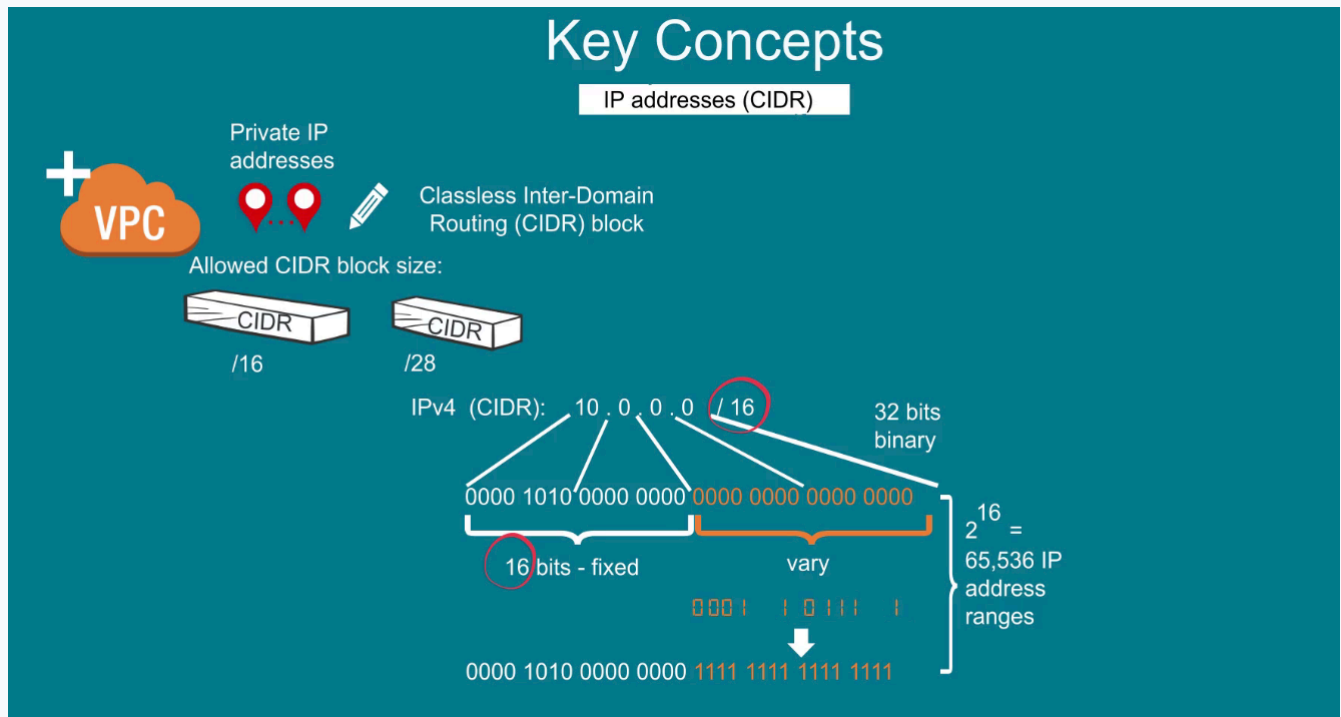
Subnet: A subnet is a portion of the VPC IP address range. Each subnet exists in one Availability Zone.

CIDR block: AWS commonly represents IPv4 network ranges with CIDR notation, such as 10.10.0.0/16 for a VPC and 10.10.0.0/24 for a smaller subnet.

Route table: A route table contains rules that direct network traffic toward a destination, such as another part of the VPC or an Internet Gateway.

Public vs private: A public subnet has routing that can support internet connectivity through an Internet Gateway. A private subnet has no direct inbound path from the internet.

CIDR blocks and IPv4 address ranges



CIDR notation describes both the starting network address and how large the address range is. For example, /16 leaves more addresses available than /24 because fewer bits are fixed as the network portion.

The important connection is: VPC address range → subnet address range → route table → reachable destinations. Security Groups then decide which of that possible traffic is actually allowed.

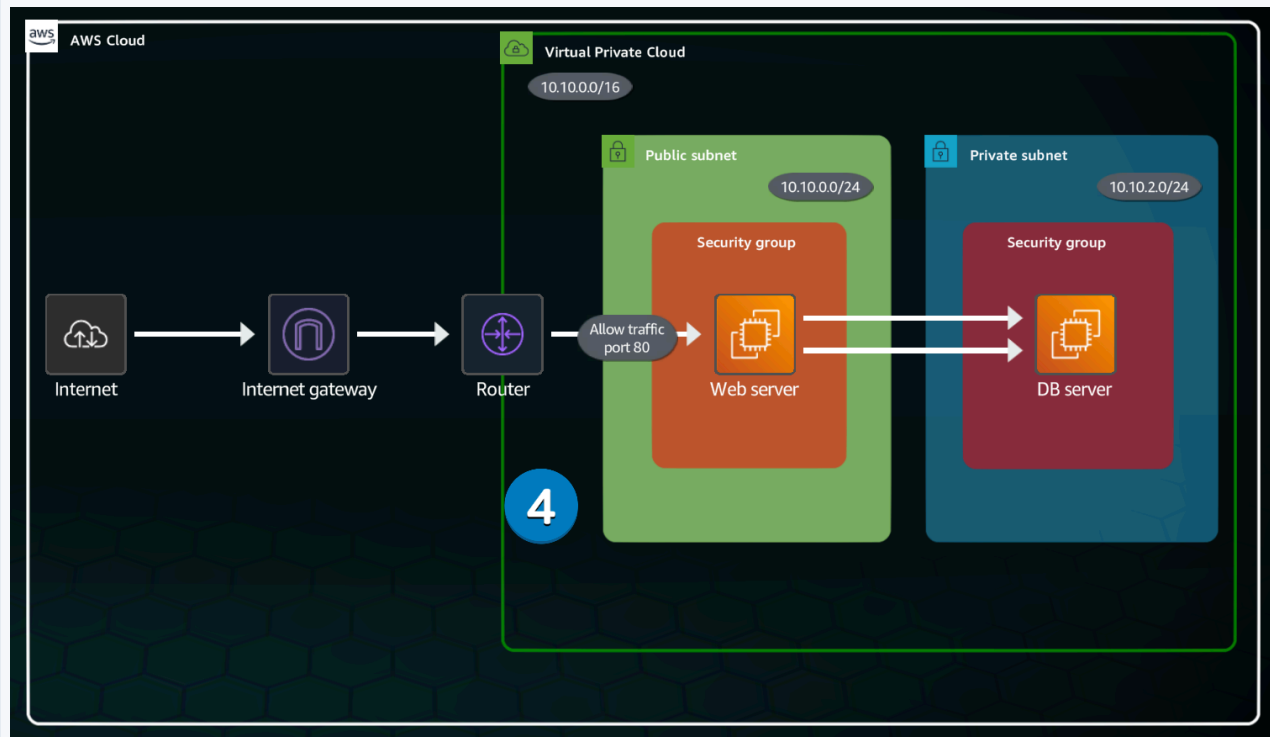
What to remember

- A VPC is the network boundary for your resources in one Region.
- A subnet belongs to one Availability Zone and uses part of the VPC IP range.
- Routes define possible traffic paths; Security Groups decide what traffic is allowed.
- Private IPs work inside the VPC. Public addressing and DNS make resources reachable in other ways.

Security Groups and Network ACLs

So far, we have decided where traffic can go using subnets, gateways and routes. The next question is whether that traffic is actually allowed. AWS gives us two important layers: Security Groups around resources such as EC2 instances, and Network ACLs (NACLs) at the subnet boundary.

Allowing web traffic into the public subnet

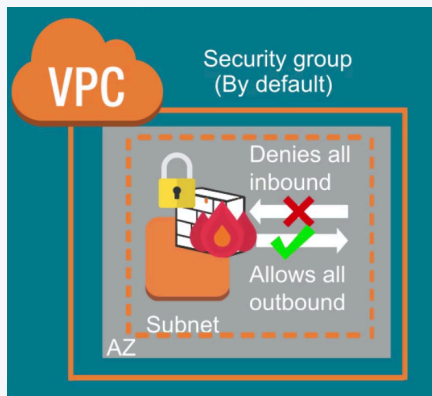


A route can lead to the web server, but the security rules still decide whether traffic such as HTTP on port 80 may reach it.

Security Groups: protecting the instance

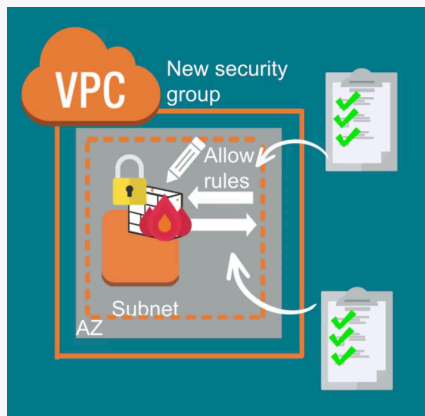
A Security Group acts as a virtual firewall for resources such as EC2 instances. It controls which inbound and outbound traffic is allowed. A new Security Group starts with no inbound rules, so inbound traffic is blocked until you explicitly allow it. It normally includes an outbound rule that allows all outbound traffic. Security Groups use allow rules rather than explicit deny rules.

Default Security Group behavior



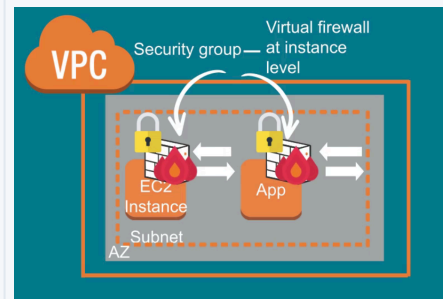
No inbound traffic is allowed unless a rule permits it.

Adding allow rules



Add only the inbound traffic the resource needs.

Instance-level firewall

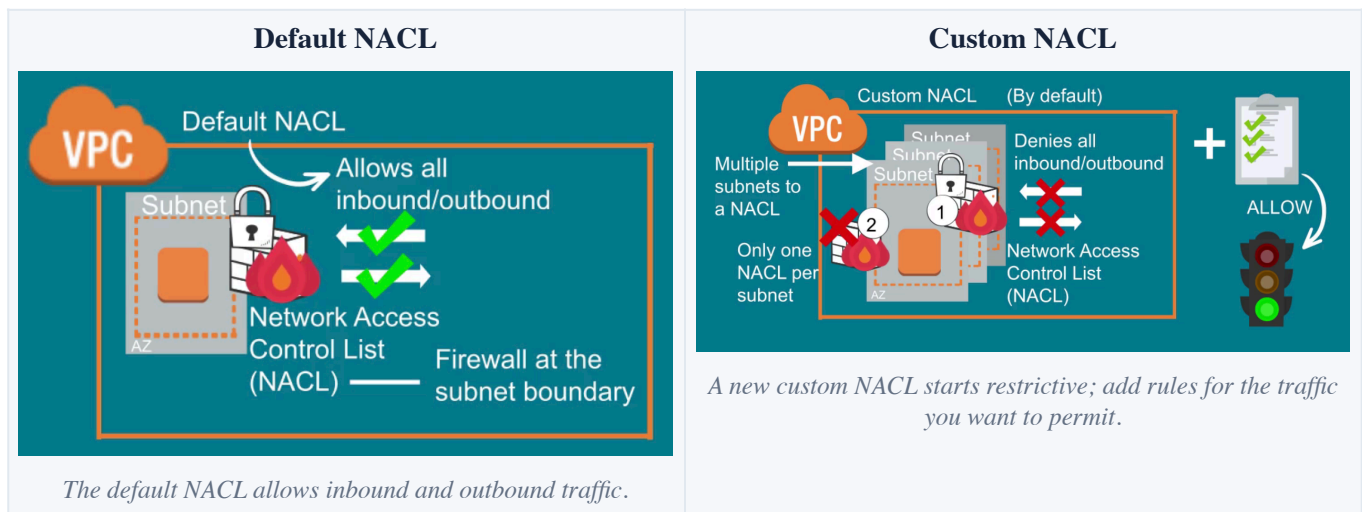


Security Groups are attached at the resource/network-interface level.

Direction	Example rule	Meaning
Inbound	HTTP · TCP · 80 · 0.0.0.0/0	Allow IPv4 web traffic from anywhere
Inbound	SSH · TCP · 22 · trusted IP	Allow SSH from a specific source
Outbound	All traffic	Allow the instance to initiate outbound traffic

Network ACLs: protecting the subnet boundary

A Network Access Control List (NACL) controls traffic at the subnet boundary. Every subnet is associated with a NACL. The default NACL that comes with a VPC allows all inbound and outbound traffic, while a new custom NACL denies traffic until matching allow rules are added.



One NACL can be associated with multiple subnets, but each subnet can be associated with only one NACL at a time. NACLs are stateless, so the necessary inbound and outbound traffic must both be permitted by rules.

How NACL rule priority works

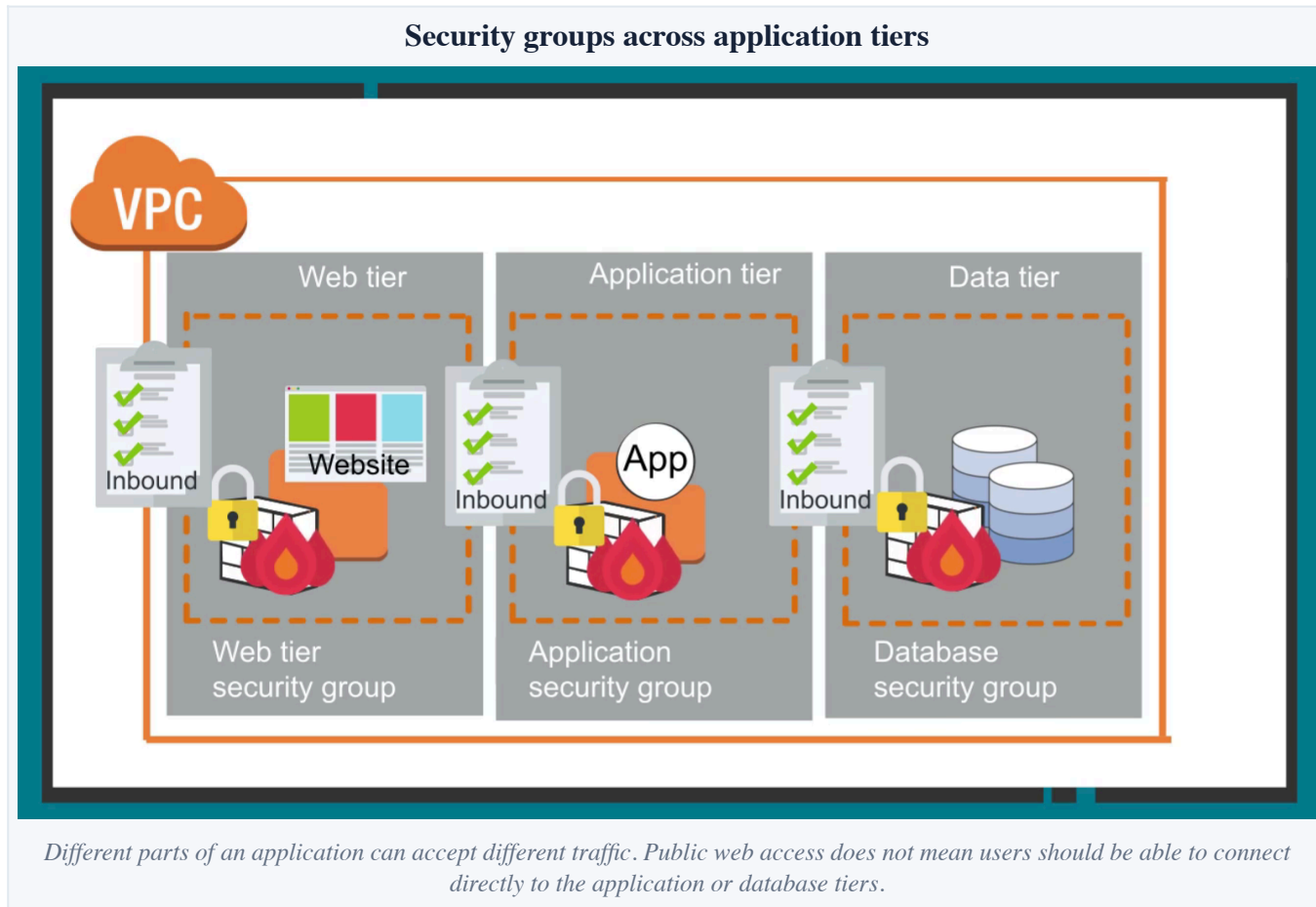
NACL rules are evaluated from the lowest rule number upward. The first matching rule decides whether the traffic is allowed or denied. The number is therefore a priority, not just a label.

Rule #	Type	Protocol	Port	Source	Action
100	HTTP	TCP	80	0.0.0.0/0	ALLOW
110	HTTPS	TCP	443	0.0.0.0/0	ALLOW
120	SSH	TCP	22	203.0.113.0/24	ALLOW
200	All traffic	All	All	0.0.0.0/0	DENY

Security Group vs NACL

	Security Group	NACL
Protects	Resource / network interface	Subnet boundary
Rules	Allow only	Allow and deny
State	Stateful	Stateless
Return traffic	Automatically recognized	Must be permitted by rules
Association	Resources / interfaces	Subnets
Rule priority	No numbered priority	Lowest rule number wins

Stateful means a Security Group remembers permitted connections, so response traffic is automatically recognized. Stateless means a NACL evaluates each direction separately, so the required inbound and outbound traffic must both be permitted.



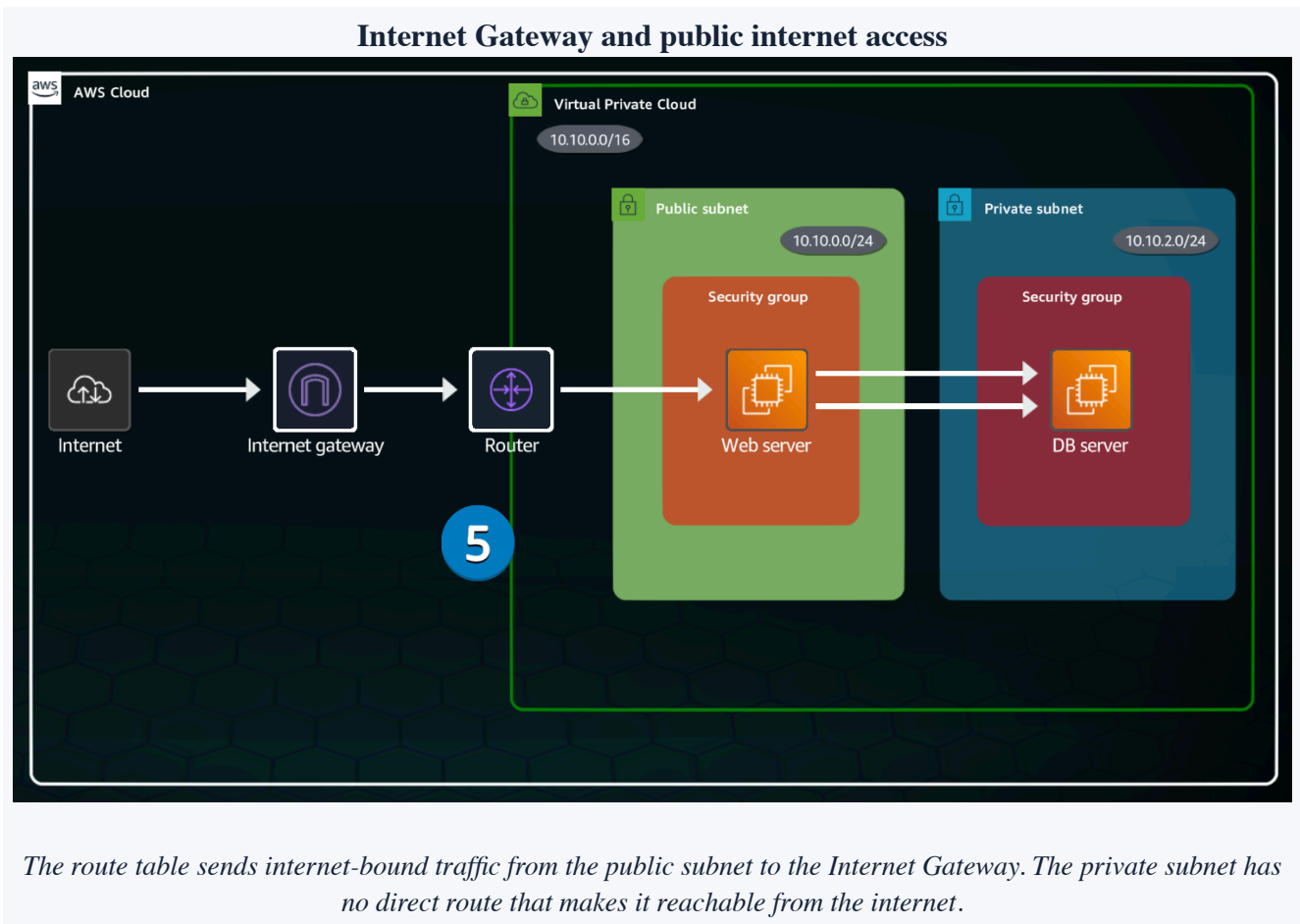
Imagine allowing customers to reach your website on port 443. That does not mean those customers should also be able to connect directly to the database. Different security boundaries let each part of the application accept only the traffic it actually needs.

What to remember

- Routes determine where traffic can go.
- Security Groups control allowed traffic at the resource level.
- NACLs control traffic at the subnet boundary.
- Security Groups are stateful and use allow rules.
- NACLs are stateless, support allow and deny rules, and evaluate numbered rules in priority order.

Internet access: Internet Gateways and NAT

Routes can point traffic toward the internet, but a VPC still needs a gateway that connects the VPC to it. Public and private subnets use that connection differently.



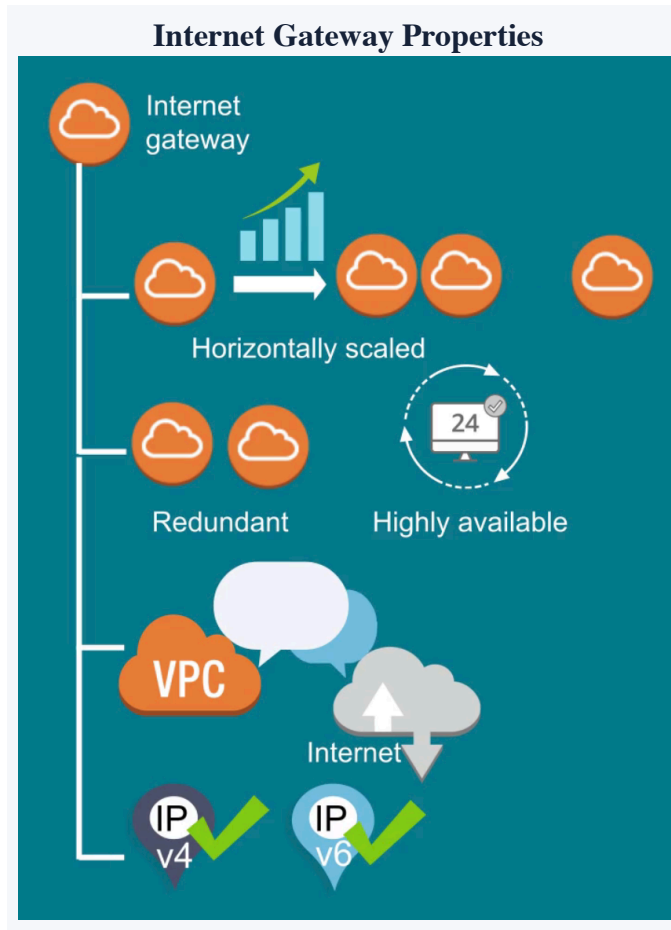
Internet Gateway: connecting the VPC to the internet

An Internet Gateway (IGW) is attached to a VPC and provides a path between that VPC and the internet. AWS describes it as horizontally scaled, redundant and highly available. It supports both IPv4 and IPv6 traffic.

For an instance in a public subnet to communicate with the internet, the pieces have to line up:

- Attach an Internet Gateway to the VPC.
- Add a route in the subnet route table that points internet-bound traffic to the Internet Gateway.
- Give the instance a public IPv4 address, an Elastic IP address, or IPv6 addressing as appropriate.
- Allow the required traffic through the Security Group and NACL.

An Elastic IP address is a static public IPv4 address. Unlike an automatically assigned public IPv4 address, it can remain associated with your AWS resources until you release it.



PUBLIC INTERNET PATH

What makes a subnet public?

A subnet is not public just because it contains a web server. Its route table must provide a path to an Internet Gateway, and the resource also needs suitable public addressing and security rules.

Internet Gateways are designed to scale and remain available while providing IPv4/IPv6 connectivity for a VPC.

Private subnets and NAT Gateways

A private subnet normally has no direct route to an Internet Gateway. That is useful for resources such as application or database servers that should not accept connections initiated directly from the internet.

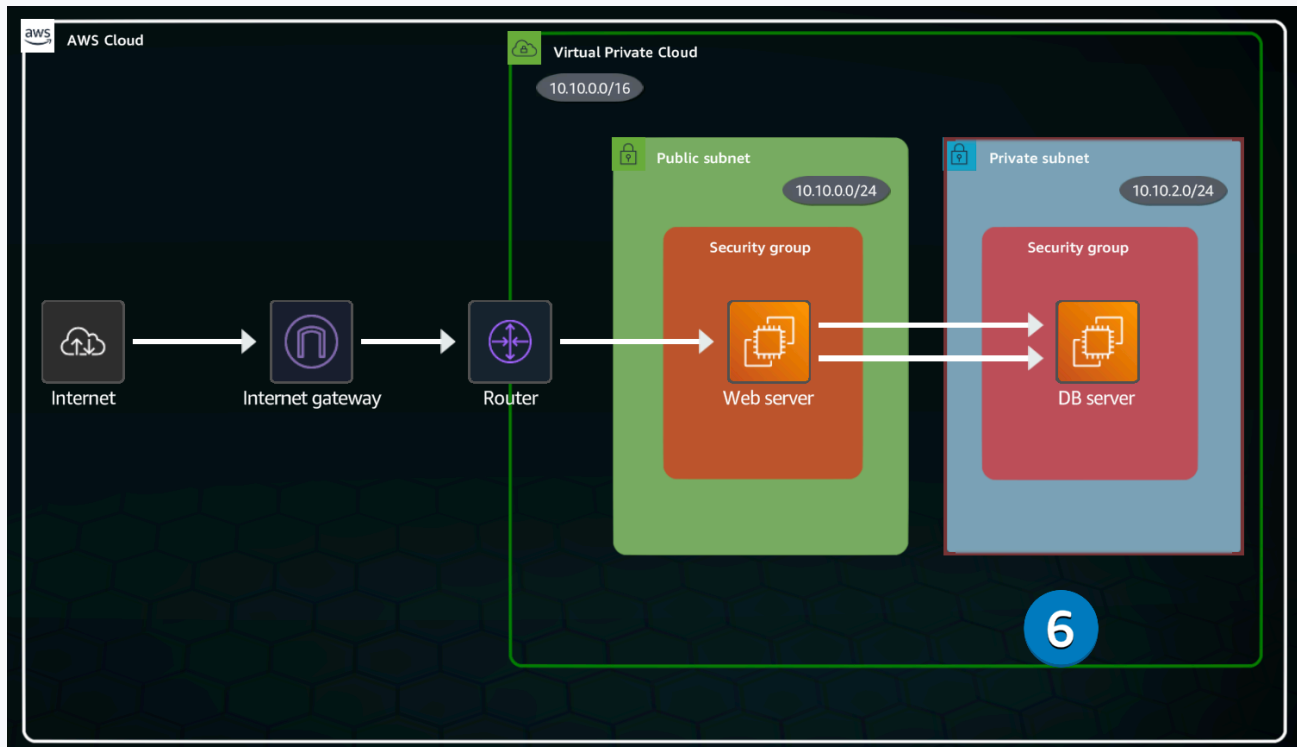
Sometimes a private instance still needs outbound internet access, for example to download updates or contact an external API. A NAT Gateway can provide that path without making the instance directly reachable from the internet.

NAT stands for Network Address Translation. In this setup, the NAT Gateway sits in a public subnet. Private instances send outbound IPv4 traffic through it, and the NAT Gateway uses its own public address when communicating with the internet. Unsolicited inbound internet connections are not forwarded back to the private instances.

Private subnet: keep the database off the internet

The database server sits in a private subnet. Its route table does not provide a direct path from the internet, so internet users cannot initiate connections to it.

Private subnet and database isolation



The public web server can receive internet traffic, while the database remains reachable only through allowed internal network paths.

This separation reduces exposure. The web server can be internet-facing because customers need to reach it, while the database only needs to accept traffic from the application side of the architecture. Security Groups and NACLs still decide which traffic is allowed; the private route configuration removes the direct internet path.

If the private database server ever needs outbound internet access, such as for updates, it can use a NAT Gateway without becoming directly reachable from the internet.

Practice lab: Build the VPC traffic path

The theory above explains what each networking component does. In the practice lab, we now follow the traffic path in the AWS console and change the configuration ourselves.

In the practice lab we will

- Explore the components that make up a Virtual Private Cloud (VPC).
- Inspect the subnet and the route table associated with it.
- Change the route for internet-bound traffic from a NAT Gateway to an Internet Gateway.
- Configure Security Group inbound rules so the web server and database accept only the traffic they need.

1. Start with the instance and its network

Amazon VPC gives you a logically isolated network inside AWS. A subnet is a smaller IP range inside that VPC and belongs to one Availability Zone. AWS resources such as EC2 instances are placed into these subnets.

Inspect the instance, public address and subnet

Instances (2) Info

Last updated less than a minute ago

Connect

Instance state

Actions

Launch Instances

Saved filter sets
Choose filter set

Find Instance by attribute or tag (case-sensitive)

< 1 >

☐	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
☐	Web Server	i-00e3e525e9f68eb5e	Running	t3.small	Initializing	us-east-1a	ec2-34-203-140-12
☐	DB Server	i-0bade82b4fc7d0f1d	Running	t3.small	Initializing	us-east-1a	ec2-13-220-97-189

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

VPC ID
 [vpc-00798d7006fca1cea \(network-concepts/VPC\)](#)

Subnet ID
 [subnet-06000dfc8a08fee78 \(WebServerSubnet\)](#)

Availability zone
 us-east-1a

Availability zone ID
 use1-az4

Outpost ID
-

We have an established VPC in the network tab... And a public IPv4 however..

Public IPv4 address

34.203.140.120 | [open address](#)

Instance state

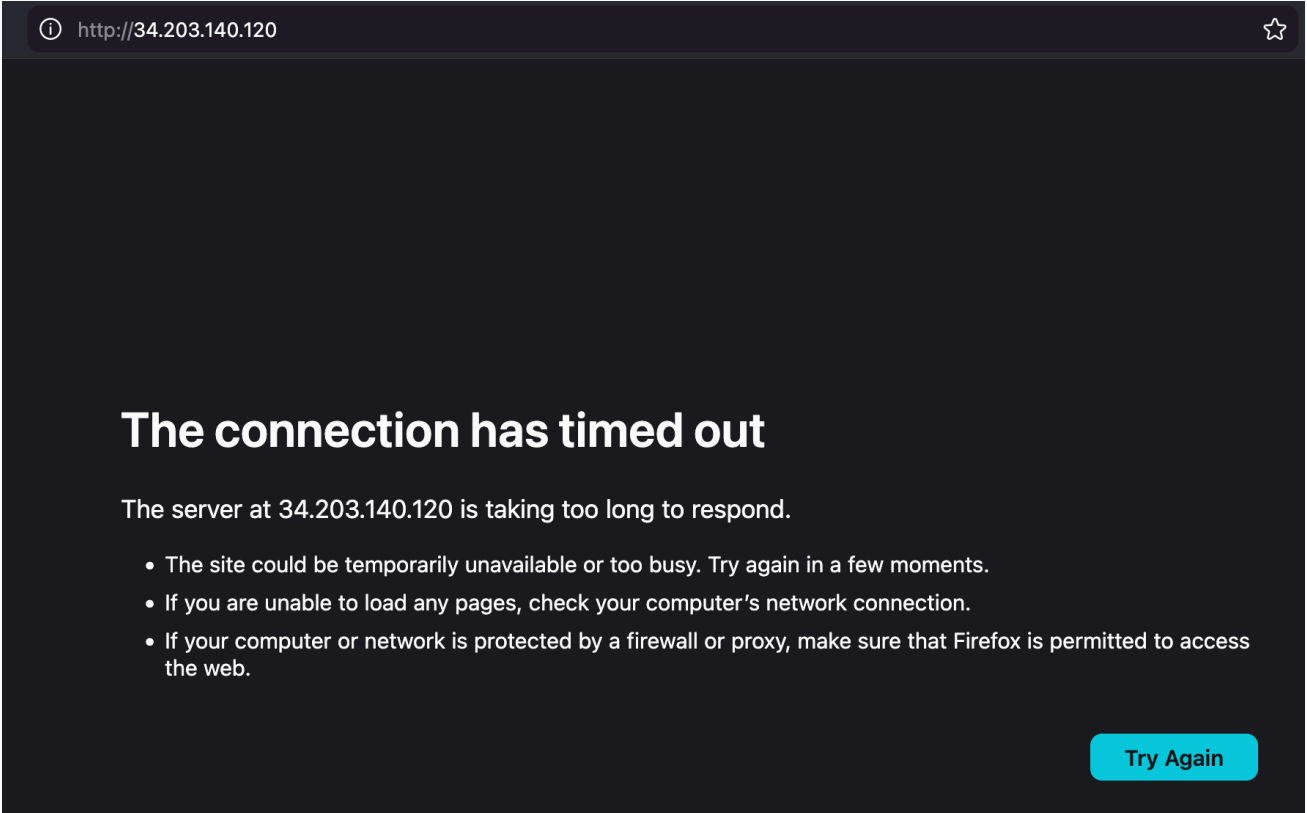
Running

Follow the instance from its overview to its public IPv4 address and the subnet where it is placed.

A public IP does not guarantee reachability

The browser connection still times out even though the instance shows a public IPv4 address. That is an important networking lesson: an address tells us how a resource can be identified, but routing and security still decide whether traffic can actually reach it.

Connection timeout



The failed connection is the clue that another part of the network path is still missing or blocking traffic.

2. Follow the subnet to its route table

From the instance, we open its subnet and inspect the route table. A route table tells AWS where traffic should go based on its destination.

Subnet and route table

Subnets (1) [Info](#)

Last updated 1 minute ago [Actions](#) [Create subnet](#)

Subnet ID: subnet-06000dfc8a08fee78 [X](#)

[Clear filters](#)

☐	Name	Subnet ID	State	VPC	Block Public...	IP
☐	WebServerSubnet	subnet-06000dfc8a08fee78	Available	vpc-00798d7006fca1cea netw...	Off	10

The web server is inside WebServerSubnet. It's routing determines where network traffic can go.

Subnets (1/1) [Info](#)

Last updated
2 minutes ago

[Actions](#)

[Create subnet](#)

Subnet ID : [subnet-06000dfc8a08fee78](#)

[Clear filters](#)

< 1 > [Settings](#)

☒	Name	Subnet ID	State	VPC	Block Public...	IP
☒	WebServerSubnet	subnet-06000dfc8a08fee78	Available	vpc-00798d7006fca1cea netw...	Off	10

subnet-06000dfc8a08fee78 / WebServerSubnet

[Details](#) | [Flow logs](#) | [Route table](#) | [Network ACL](#) | [CIDR reservations](#) | [Sharing](#) | [Tags](#) | [Related resources](#)

Route table: [rtb-07b552eb902d36a05](#) / [RouteTable2](#)

[Edit route table association](#)

Routes (2)

< 1 > [Settings](#)

Destination	Target
10.10.0.0/16	local
0.0.0.0/0	nat-0692196be634de0ad

*10.10.0.0/16 is local, keeps VPC traffic internal
0.0.0.0/0 is NAT Gateway which provides outbound internet access.*

Route tables (1) [Info](#)

Last updated
5 minutes ago

[Actions](#)

[Create route table](#)

Route table ID : [rtb-07b552eb902d36a05](#)

[Clear filters](#)

< 1 > [Settings](#)

☐	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
☐	RouteTable2	rtb-07b552eb902d36a05	2 subnets	–	No	vpc-00798d7006

WebServerSubnet uses RouteTable2, which controls where its traffic is directed.

Route tables (1/1) Info

Last updated 7 minutes ago

Actions
Create route table

Route table ID : rtb-07b552eb902d36a05
Clear filters

1

☒	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
☒	RouteTable2	rtb-07b552eb902d36a05	2 subnets	-	No	vpc-00798d7006

rtb-07b552eb902d36a05 / RouteTable2

Details
Routes
Subnet associations
Edge associations
Route propagation
Tags
Related resources

Routes (2)

Both
Edit routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	nat-0692196be634de0ad	Active	No	Create Route
10.10.0.0/16	local	Active	No	Create Route Table

The default route currently points to a NAT Gateway.

To make the web server publicly reachable, we will change this to the Internet Gateway.

Destination	Target	Meaning
10.10.0.0/16	local	Traffic for addresses inside the VPC stays inside the VPC.
0.0.0.0/0	NAT Gateway	All other IPv4 traffic is sent toward the NAT Gateway.

The destination 0.0.0.0/0 is CIDR notation for all IPv4 addresses. Because the more specific 10.10.0.0/16 route is checked for VPC traffic, 0.0.0.0/0 effectively becomes the fallback route for destinations outside the VPC.

Why the NAT route does not make the server public

A NAT Gateway is mainly used so resources in private subnets can initiate IPv4 connections to the internet without accepting unsolicited connections initiated from the internet.

Private EC2 → NAT Gateway → Internet: yes

Internet → NAT Gateway → Private EC2: no

So the NAT route can give the server outbound internet access, but it does not create the inbound path your browser needs.

3. Replace the NAT route with an Internet Gateway

In this lab we remove the NAT Gateway route and point 0.0.0.0/0 to the VPC's Internet Gateway instead. This changes the subnet's internet path.

Change the internet route

Edit routes: rtb-07b552eb902d36a05 / RouteTable2

Destination	Target	Status	Propagated	Route Origin
10.10.0.0/16	local	✓ Active	No	CreateRouteTable
	Q local			

[Add route](#)

[Cancel](#) [Preview](#) [Save changes](#)

*With the removal of 0.0.0.0/0, the NAT Gateway is no more.
The NAT Gateway lets the server initiate connections out to the internet.
It does not let internet users initiate connections to the web server.*

*Because our goal is to make the web server publicly reachable, this route does not provide
the type of internet access we need.*

Edit routes: rtb-07b552eb902d36a05 / RouteTable2

Destination	Target	Status	Propagated	Route Origin
10.10.0.0/16	local	✓ Active	No	CreateRouteTable
	Q local			
Q 0.0.0.0/0	Internet Gateway	-	No	CreateRoute
	Q igw-027b427dfb94e936e			

[Add route](#)

[Cancel](#) [Preview](#) [Save changes](#) [Remove](#)

Add 0.0.0.0/0 Internet Gateway, giving the subnet a direct route to and from the internet.

Routes (2) [Both](#) [Edit routes](#)

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-027b427dfb94e936e	✓ Active	No	Create Route
10.10.0.0/16	local	✓ Active	No	Create Route Table

*The Internet Gateway route is now Active.
The web server can be publicly reachable when its public ip and security rules also allow it.*

Destination	Target	Meaning
10.10.0.0/16	local	Keep VPC traffic inside the VPC.
0.0.0.0/0	Internet Gateway	Send unmatched IPv4 traffic toward the internet.

You can mentally read this as: inside our VPC, keep it local; anything else, send it toward the Internet Gateway.

The Internet Gateway gives the VPC a direct internet path, but the instance still needs suitable addressing and security rules. For IPv4, that normally means a public IPv4 address or Elastic IP. An Elastic IP is a static public IPv4 address that you control in your AWS account.

There is no hourly charge for the Internet Gateway itself, although normal AWS data-transfer charges can still apply. NAT Gateways, by contrast, have their own usage charges while provisioned.

4. Open the Security Group

Routing gets traffic to the right network path. The Security Group is the next checkpoint: it decides which traffic is allowed to reach the instance.

Navigate to the instance Security Group

Instances (1/2) Info
Last updated 32 minutes ago
Connect
Instance state
Actions
Launch instances

Saved filter sets
Choose filter set

Find Instance by attribute or tag (case-sensitive)

☐	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
☒	Web Server	i-00e3e525e9f68eb5e	Running	t3.small	Initializing	us-east-1a	ec2-34-203-140-12
☐	DB Server	i-0bade82b4fc7d0f1d	Running	t3.small	Initializing	us-east-1a	ec2-13-220-97-189

i-00e3e525e9f68eb5e (Web Server)

Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags

▼ Security details

IAM role
ec2_app_role-19a4e420

Owner ID
408687975267

Launch time
Thu Aug 27 2026 13:43:42 GMT+0200 (Central European Summer Time)

Security groups
sg-0c8f85c5a4a66a9d2 (WebServerSecurityGroup)

*The web server uses WebServerSecurityGroup.
Security Groups control which traffic can enter or leave the instance.*

sg-0c8f85c5a4a66a9d2 - WebServerSecurityGroup

Actions

Details

Security group name

WebServerSecurityGroup

Security group ID

sg-0c8f85c5a4a66a9d2

Description

Security Group for Web Server

VPC ID

vpc-00798d7006fca1cea

Owner

408687975267

Inbound rules count

0 Permission entries

Outbound rules count

3 Permission entries

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules

Manage tags

Edit inbound rules

Search

< 1 >

	Name	Security group rule ID	IP version	Type	Protocol	Port range
No security group rules found						

The Security Group currently has **no inbound rules**, so incoming connections are blocked.

Next up is allowing the web traffic to hit the server.

Allow HTTP to the public web server

We add an inbound rule for HTTP on TCP port 80 with source Anywhere-IPv4. This explicitly allows any IPv4 address to attempt an HTTP connection to the web server.

Add the HTTP inbound rule

Edit inbound rules

Info

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules

Info

Security group rule ID	Type	Info	Protocol	Info	Port range	Source	Info	Description - optional	Info
-	HTTP		TCP		80	Anyw...			

Add rule

0.0.0.0/0

Delete

Rules with source of 0.0.0.0/0 or :::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Cancel

Preview changes

Save rules

AWS warns about broad sources because 0.0.0.0/0 exposes that port to the public internet.

For a public web server, allowing HTTP or HTTPS from the internet can be expected. The same broad source would usually be inappropriate for sensitive management ports or a database.

5. Allow only the web tier to reach MySQL

The database should not accept MySQL connections from the whole internet. Instead, its Security Group can allow TCP port 3306 with the web server's Security Group as the source.

AWS Cloud • Part II

Restrict database access to the web Security Group

Inbound rules | **Outbound rules** | Sharing | VPC associations | Related resources - new | Tags

Outbound rules (3)



Manage tags

Edit outbound rules

Q Search

< 1 > ⚙

☐	Name ▾	Security group rule ID ▾	IP version ▾	Type ▾	Protocol ▾	Port range ▾
☐	-	sgr-0ef0695a58164f1f4	IPv4	HTTPS	TCP	443
☐	-	sgr-092544613b7dae35f	-	MYSQL/Aurora	TCP	3306
☐	-	sgr-02d8ecea36ecdfd95	IPv4	HTTP	TCP	80

Outbound rules control where the web server is allowed to send traffic.

Edit outbound rules [Info](#)

Outbound rules control the outgoing traffic that's allowed to leave the instance.

Outbound rules [Info](#)

Security group rule ID	Type Info	Protocol Info	Port range Info	Destination Info	Description - optional Info
sgr-0ef0695a58164f1f4	HTTPS ▾	TCP	443	Custom ▾ Q	Allow HTTPS traffic for S3, S3 Delete
sgr-092544613b7dae35f	MYSQL/Aurora ▾	TCP	3306	Custom ▾ Q	Allow Web Server to connect Delete
sgr-02d8ecea36ecdfd95	HTTP ▾	TCP	80	Custom ▾ Q	Allow Web Server to connect to DB Server on port 3306 Delete

Add rule

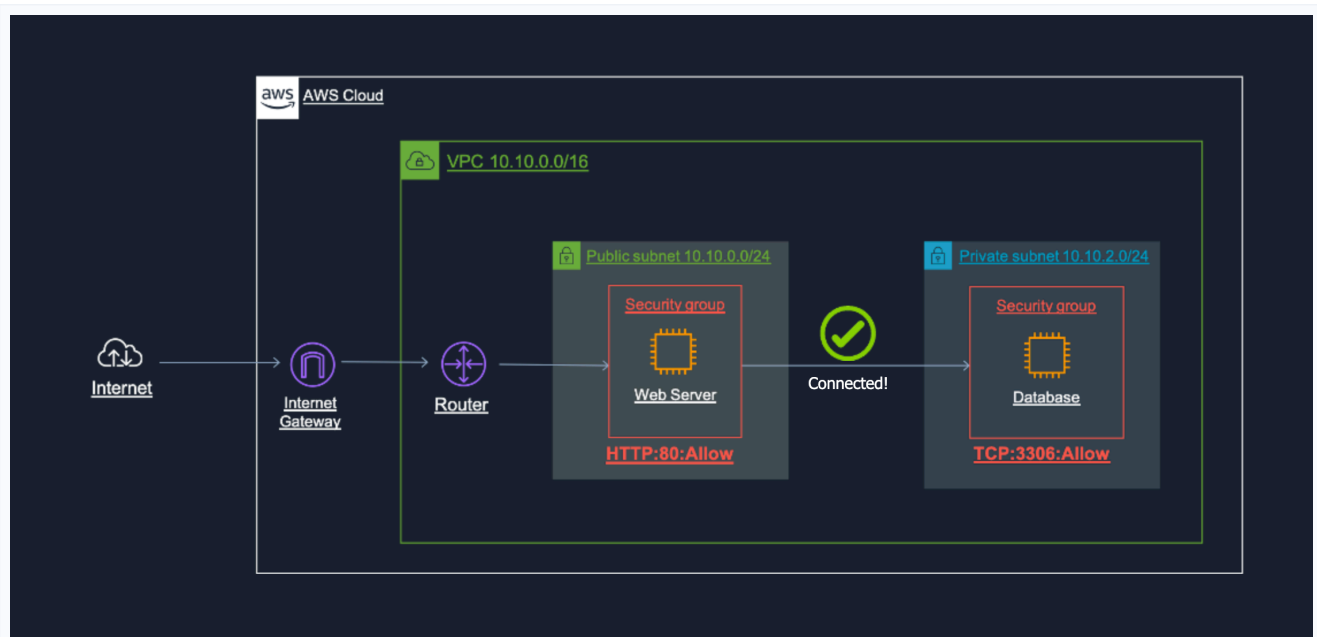
⚠ Rules with destination of 0.0.0.0/0 or ::/0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific known IP addresses.

Cancel

Preview changes

Save rules

Keep only the connections the web server needs, including MySQL on TCP 3306.



*Port 3306 is allowed specifically to the database Security Group.
The web server can now communicate with the database without exposing the DB to everyone.*

The lab's final mental model

The lab brings the networking pieces together. No single setting makes a server reachable. The path has to work from end to end.

A VPC is our private AWS network, divided into subnets with defined IP ranges, while route tables direct traffic through gateways such as an Internet Gateway or NAT Gateway, and Security Groups and NACLs act as firewalls controlling what traffic is allowed.